■ Developer Mock Test Results

Test ID:	d161c46a-2b5e-47...	**Student ID:**	5908
Test Type:	Developer Assessment	**Date:**	2026-01-20 10:40:03
Overall Score:	10/25 (40.0%)	**Performance:**	■ Average
Warnings:	0/3	**Status:**	Completed ■

■ Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	7/10	70.0%	■ Pass
■ MCQ/Theory	3/10	30.0%	■■ Needs Work
■ Coding	0/5	0.0%	■■ Needs Work

■ Detailed Question Review

■ APTITUDE SECTION (7/10 - 70.0%)

■ Q1. Two people, A and B, can do a piece of work in 10 days and 15 days respectively. How many days will it take for A and B to do the work together?

Your Answer: 6 days

■ *Correct! Your logical reasoning is on point.*

■ Q2. A mixture is made of two liquids A and B in the ratio 2:3. If the quantity of liquid A is 24 liters, what is the quantity of liquid B?

Your Answer: 36 liters

■ *Correct! Your logical reasoning is on point.*

■ Q3. The average of 10 numbers is 15. If one of the numbers is 20, what is the sum of the remaining 9 numbers?

Your Answer: 130

■ *Well done! You solved this problem correctly.*

■ Q4. There are five houses, each painted a different colour: blue, green, red, white, and yellow. Each house is occupied by a person of a different nationa...

Your Answer: Green

Correct Answer: Blue

■ *The correct answer is Blue because the Canadian lives in the first house, and since the American lives in the red house, the Canadian cannot live in the red house, and by process of elimination, the first house must be blue. The user likely made a mistake by not carefully considering the position of the Canadian's house and the constraints provided by the other clues, such as the American living in the red house. Don't worry, it's all part of the learning process, and with practice, you'll become more skilled at using logic and deduction to solve puzzles like this one.*

■ Q5. What is the next number: 3, 6, 9, 12, 15, ?

Your Answer: 20
Correct Answer: 18
■ *The correct answer is 18 because the pattern in the sequence is adding 3 to the previous number, so the sequence goes 3, 6, 9, 12, 15, 18. It seems you were close, but might have misidentified the pattern, possibly thinking it was an increase of a different amount or a different type of sequence, which led you to choose 20. Don't worry, it's all part of the learning process, and recognizing patterns takes practice, so keep going and you'll get better at it.*

■ Q6. What is the next number: 2, 6, 12, 20, 30, ?

Your Answer: 42
■ *Excellent! Your mathematical approach was correct.*

■ Q7. If price increases from Rs.200 to Rs.250, what is the percentage increase?

Your Answer: 25%
■ *Excellent! Your mathematical approach was correct.*

■ Q8. A shopkeeper buys 100 items at Rs.10 each and sells 50 items at Rs.15 each. What is the percentage profit?

Your Answer: 50%
Correct Answer: 25%
■ *The correct answer is 25% because the shopkeeper's total cost is Rs.1000 for 100 items, and selling 50 items at Rs.15 each gives a revenue of Rs.750, but since the unsold items are still part of the total cost, the profit is calculated on the total cost, resulting in a 25% profit. The user likely made a mistake by calculating the profit only on the sold items, which would indeed give a 50% profit, but this doesn't account for the remaining unsold items. Don't worry, it's an easy mistake to make, and with practice, you'll become more accurate in calculating profits and losses.*

■ Q9. A pipe can fill a tank in 6 hours. Another pipe can empty the tank in 8 hours. How long will it take to fill the tank if both pipes are open?

Your Answer: 24 hours
■ *Correct! Your logical reasoning is on point.*

■ Q10. A mixture is made of two liquids A and B in the ratio 3:5. If the quantity of liquid A is 36 liters, what is the quantity of liquid B?

Your Answer: 60 liters
■ *Excellent! Your mathematical approach was correct.*

■ MCQ SECTION (3/10 - 30.0%)

■ Q1. What is the primary benefit of using multithreading in Python?

Your Answer: Enhanced memory management
Correct Answer: Increased responsiveness and performance in I/O-bound applications
■ *The correct answer is increased responsiveness and performance in I/O-bound applications because multithreading in Python allows your program to perform other tasks while waiting for input/output operations to complete, making it more efficient. The user's answer, enhanced memory management, is likely incorrect because multithreading doesn't directly improve memory management, and this concept is more related to other programming techniques. Don't worry, it's a common misconception, and with practice, you'll become more familiar with the benefits of multithreading in Python.*

■ Q2. What is the purpose of the threading module in Python?

Your Answer: To handle file operations and data management
Correct Answer: To create and manage threads using thread objects and methods
■ *The correct answer is to create and manage threads using thread objects and methods because the threading module in Python provides a way to run multiple threads or flows of execution concurrently, which is useful for tasks that require simultaneous execution. It seems you may have mistaken the purpose of the threading module with other Python modules that handle file operations and data management, such as the os or pandas modules. Don't worry, it's an easy mistake to make, and now you know the correct purpose of the threading module, so you can use it effectively in your Python programming journey.*

■ **Q3. What is the correct order of steps to execute a thread in Python?**

Your Answer: Import the threading module, define a function to run in a thread, create a Thread object

■ *Correct! You have a good understanding of this concept.*

■ **Q4. What is the purpose of using locks in Python threading?**

Your Answer: To prevent data corruption and ensure thread safety

■ *Correct! You have a good understanding of this concept.*

■ **Q5. What is the primary limitation of the Global Interpreter Lock (GIL) in Python?**

Your Answer: It prevents the use of multithreading in Python

Correct Answer: It limits the performance of CPU-bound multithreading

■ *The correct answer is right because the Global Interpreter Lock (GIL) in Python indeed limits the performance of CPU-bound multithreading, as it prevents multiple native threads from executing Python bytecodes at once, which can hinder performance in computationally intensive tasks. The user's mistake likely stems from misunderstanding the GIL's role, as it does not prevent the use of multithreading altogether, but rather imposes a limitation on its effectiveness in certain scenarios. By recognizing this distinction, you can better understand how to optimize your Python code for multithreading, and explore alternative approaches, such as multiprocessing or asynchronous programming, to achieve improved performance.*

■ **Q6. What is the primary purpose of file handling in Python?**

Your Answer: To manage threads and processes

Correct Answer: To store, retrieve, and process data from external files

■ *The correct answer is to store, retrieve, and process data from external files because file handling in Python allows you to interact with files outside of your program, enabling data persistence and exchange. It seems you may have confused file handling with another concept, possibly multiprocessing, which is used to manage threads and processes, but is a separate topic in Python. Don't worry, it's an easy mistake - just remember that file handling is all about working with external files to store and retrieve data, and you'll be on the right track.*

■ **Q7. What is the purpose of the 'a' mode in Python file handling?**

Your Answer: To read data from a file

Correct Answer: To add data to the end of a file

■ *The 'a' mode in Python file handling is used to add data to the end of a file, making it useful for appending new information without overwriting existing content. The user's answer was likely incorrect because they may have confused the 'a' mode with the 'r' mode, which is used for reading data from a file. Don't worry, it's an easy mistake to make, and now you know the correct purpose of the 'a' mode, so you can use it effectively in your future programming endeavors.*

■ **Q8. What is the correct step to read data from a file in Python?**

Your Answer: Open the file using the open() function, read data using read(), close the file

■ *Well done! Your knowledge is solid here.*

■ **Q9. What is the purpose of the csv and json libraries in Python?**

Your Answer: To implement object-oriented programming principles

Correct Answer: To simplify reading and writing structured data

■ *The csv and json libraries in Python are used to simplify reading and writing structured data, making it easier to work with comma-separated values and JavaScript Object Notation files. The user's answer was incorrect because object-oriented programming principles are a broader concept in Python, and while important, they aren't directly related to the purpose of these libraries. Don't worry, it's an easy mistake - the key is to remember that csv and json libraries are specifically designed to handle structured data, making data exchange and storage more efficient.*

■ **Q10. What is the purpose of using try and except blocks in Python file handling?**

Your Answer: To prevent data corruption and ensure thread safety

Correct Answer: To handle errors and exceptions during file operations

■ *The correct answer is to handle errors and exceptions during file operations because try and except blocks in Python allow you to catch and manage errors that may occur when working with files, such as file not found or permission errors. The user's answer was likely incorrect because preventing data corruption and ensuring thread safety are important considerations in file handling, but they are not the primary purpose of using try and except blocks. By using try and except blocks, you can write more robust and reliable code that can handle unexpected errors and provide a better user experience, so don't be discouraged by the mistake - keep practicing and you'll become proficient in using these blocks to* *handle file operations effectively.*

■ CODING SECTION (0/5 - 0.0%)

■ Q1. Write a Python function to calculate the average grade of a student given their scores in three subjects: math, science, and English. **Input:** A dic...

Your Answer: SSS

Correct Answer: None

■ *The correct answer involves creating a Python function that takes a dictionary of subject scores, sums the scores, and divides by the number of subjects to calculate the average grade. The user's answer, "SSS", doesn't provide any functional code, suggesting they may have misunderstood the question or not attempted to write a Python function. Don't worry, it's a great opportunity to learn and practice writing functions in Python, and with a bit of effort, you can create a working solution to calculate the average grade.*

■ Q2. Write a Python program to count the number of lines in a given text file. **Input:** The name of the text file (e.g., "example.txt") **Output:** The n...

Your Answer: S

Correct Answer: None

■ *The correct answer involves using Python's built-in file handling functions to open and read the file, then counting the number of lines, which can be achieved with a simple loop or the len() function in combination with readlines(). The user's answer, "S", seems unrelated to the task, suggesting they may have misunderstood the question or failed to provide a code-based solution. Don't worry, it's a great opportunity to learn and practice Python file handling, and with a bit of practice, you'll be able to write a correct program to solve this problem.*

■ Q3. Write a Python class to manage bank account transactions. The class should have methods to deposit, withdraw, and check the balance. **Input:** Initia...

Your Answer: dd

Correct Answer: None

■ *The correct answer involves creating a Python class with methods to manage bank account transactions, which is the most appropriate way to solve this problem. The user's answer, "dd", seems to be incomplete and unrelated to the problem, suggesting that they may have misunderstood the question or didn't know where to start. Don't worry, it's okay to make mistakes - let's try breaking down the problem and creating a simple class with deposit, withdraw, and balance check methods to achieve the desired outcome.*

■ Q4. Write a Python program to download multiple files concurrently using threads. **Input:** A list of file URLs (e.g., `["http://example.com/file1.txt", ...

Your Answer: dd

Correct Answer: None

■ *The correct answer utilizes Python's threading capabilities to download multiple files concurrently, which is an efficient approach for tasks that involve waiting, such as downloading files. The user's answer, "dd", seems unrelated to the task, suggesting they may have misunderstood the question or failed to provide a relevant code snippet. Don't worry, this is a great opportunity to learn about threading in Python and practice writing concurrent programs to achieve tasks like this.*

■ Q5. Write a Python program to convert a CSV file to a JSON file. **Input:** The name of the CSV file (e.g., "data.csv") **Output:** The name of the result...

Your Answer: dd

Correct Answer: None

■ *The correct answer involves using Python's built-in csv and json modules to read the CSV file and write its data to a JSON file, but since no code was provided, we'll have to start from scratch. The user's answer, "dd", seems unrelated to the task, suggesting they may have misunderstood the question or not attempted to write a Python program. Let's try again, and I'll guide you through the process of converting a CSV file to a JSON file using Python.*

■ Recommendations

Areas that need improvement:

- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.